

# Manual

## Joint Production HLS-KUP 8.2

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# 1 Overview Joint Production

## Purpose

The function package HLS-KUP 8.2 provides the option to link operations in the HYDRA Shop Floor Scheduling (HLS) and to plan them as one.

## Implementation notes

You use the function package HLS-KUP 8.2 if you want to integrate in the planning the parallel processing of different operations at one workplace. Example: This scenario is frequent with manufacturing companies in injection molding where you often produce left and right parts with one tool. Here, a production order with different article numbers exists for the left and the right part.

If you want to plan these operations as one and log the operations on as one to the terminal, you must link the operations. You link the operations in the application "Graphic Planning".

- You can link operations of different production orders to plan them as one in the HYDRA Shop Floor Scheduling
- You can plan the same tool for linked operations without conflict
- You make one posting for linked operations on the terminal

## Integration

The function package HLS-KUP 8.2 is integrated into the HYDRA Shop Floor Scheduling. You can log on, log off and interrupt linked operations on the terminal as one.

## Features

- Function to link operations in the graphic planning board. The linked operations are planned in parallel (identical planned start).
- If you link operations, you can plan the simultaneous production of left and right parts, for example.
- Configuration of the maximum number of operations that you can link
- Visualization of the linked operations
- Functions to schedule, cancel and replan linked operations
- You can log on, log off and interrupt the linked operations as one on the shop floor terminal if you use the AIP.

## 2 Joint Production

### Overview

|                        |                                                                     |
|------------------------|---------------------------------------------------------------------|
| HYDRA menu             | Production control → Preparations for production → Graphic planning |
| FEDRA menu             | Detailed Scheduling → Planning → Graphic planning                   |
| Transaction code       | grap                                                                |
| Function authorization | grap                                                                |

### Purpose

You use the joint production if you want to link operations and plan them for a workplace as one operation. Linked operations have an identical planned start.

### Integration

You can use the graphic planning board to plan capacities for machines and workplaces in your production department.



With operations linked via the "joint production" function, you cannot use the following functions:

- the "coil cutting" function (MPL)
- the function to build a merged operation (BDE) (only applies if HYDRA is used).

### Requirements

Proceed as described below in order to display the most up-to-date planning in the graphic planning board:

- You have created orders required for planning on the system (production orders, maintenance orders, project orders) or you have downloaded these orders from a higher-level system.
- You have defined the responsibilities regarding order planning in your company and determined which workplace/machine groups are used for planning. Based on these initial steps, you have defined the necessary [planning profiles](#).

You require the respective authorizations to be able to [change the planning](#) in the graphic planning board.

### Joint production – settings

In the settings of the *Graphic Planning*, you must configure the specifications that are used to link operations. Make the settings for joint production in tab *Operation links*. The settings listed in the following can be edited.

- *Max. number of linked OPs:*  
Enter the maximum number of operations that can be linked.
- *Linked OPs visualization:*
  - o *Only show master OP*  
Only the master operation is displayed. The included operations that have been linked are not displayed.
  - o *Show master OP with icon*  
Only the master operation is displayed. The included operations that have been linked are not displayed. The master operation is identified using a symbol.
  - o *Show all operations*  
All operations linked are shown.
  - o *Show all operations with icon*  
All operations linked are shown. Each of the operations linked is identified using a symbol.
- *Allow linking of planned operations:*  
Specify if you are allowed to link operations that have already been scheduled. If you do not enable the option, you can only link operations of the pool of groups.
- *Do not adjust bar length:*  
If you enable the option, the bar length of the individual operations linked does not change. If you do not enable the option, the bar lengths of all operations linked are changed to the bar length of the operation with the longest bar.

## Joint production – Linking and unlinking

### Linking operations

You require the function authorization "op.cmbbuild" to link operations.

Select the operations that you want to link. You can select the operations in the planning board or in the tables (pool of groups, pool of workplaces). The relevant configurations in the system settings specify whether you can link operations that have already been scheduled.

After having selected the operations, right-click to open the context menu of one of the selected operations. Select the entry "Link operations". The system then links the selected operations. The system checks if the maximum number of operations that can be linked is respected. The system uses the configuration made in the system settings to this end.

The configurations made in the system settings also specify how the bars of the linked operations are displayed and if the bar lengths change.

If you link several operations, the mother operation does not change the position (planned start and planned end on workplace or group level). The other operations linked are based on the mother operation. These operations now have the same planned start as the mother operation. The mother operation is the operation with the longest bar length.



All operations linked must have the same tool stored for the operation or no tool at all.



If you have linked operations and you want to add further operations, you must first cancel the existing operation links and unlink the operations. You cannot add further operations to operations that are already linked.

### Unlinking operations

You require the function authorization "op.cmbdissolv" to unlink operations.

If you want to unlink operations, open the context menu of one of the linked operations and select "Unlink operations". You are then free to plan and link the operations as required.



If the bar lengths of the unlinked operations have been changed during linking, the original bar lengths are only restored after a new "Request data".

### Joint production – Perform planning actions

You can plan, deallocate and replan linked operations. To perform planning actions, you can use the functions you are authorized for. The system performs all checks of planning actions for the master operation only (for example, conflict checks or sequence of operations assigned to a workplace). The system does not perform any checks for the operations linked with the master operation.



The terms "master operation" and "mother operation" are used synonymously.

### Joint production – Posting

The system can process postings of linked operations in parallel. If required, you can configure for the order type that the operations are automatically logged on, logged off and interrupted in parallel (tab "Processing", field "Parallel order posting").

If several operations are linked and you interrupt/log off or post a part quantity, the collected quantities are only posted for the selected operation. The postings for the operations linked to the master operation must be made separately, if required.



If the *Parallel order posting* is activated, this does not mean that, on the AIP, resources are automatically logged on for all operations linked. For each operation, you must manually log on input batches, tools, etc.